

1 Summary

Model (Run-name): model1 (test)

Iteration: 37

TAC: 154893.30355730935

Took: 3.9788014888763428s

Total Time: 84.82882809638977s

Epsilons:

- ε_{LMTD} : 1e-06
- $\varepsilon_{Balancing}$: 1e-06
- ε_{Area} : 1e-06
- ε_{Beta} : 1e-06

Gurobi Tolerances:

- IntFeasTol: 1e-05
- FeasibilityTol: 1e-06
- OptimalityTol: 1e-06

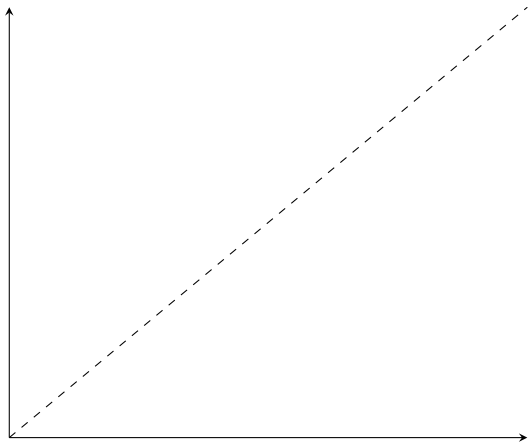
Gamsworld Solution: http://www.gamsworld.org/minlp/minlplib2/html/heatexch_gen1.html

2 Results

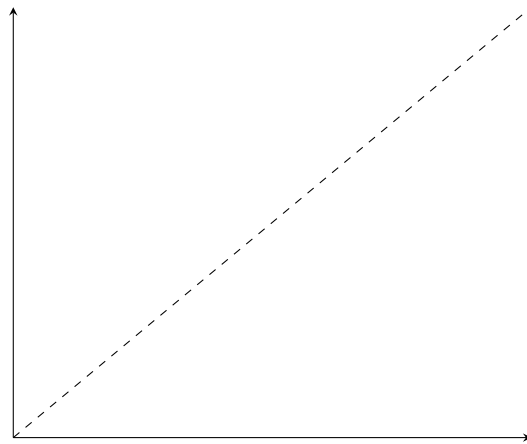
Variable	Value	Variable	Value	Variable	Value
$z_{1,1,1}$	1.0	$RecLMTD_{1,1,1}$	0.05232330513567375	$b_{1,1,1}^{H,out}$	894.499520225555
$z_{1,2,2}$	1.0	$RecLMTD_{1,2,2}$	0.01758266884067111	$b_{1,2,2}^{H,out}$	3398.463420630389
$z_{2,1,2}$	1.0	$RecLMTD_{2,1,2}$	0.02937517129217052	$b_{2,1,2}^{H,out}$	9365.377833769548
$z_{cu,1}$	1.0	$RecLMTD_{cu,1}$	0.014567639482481746	$b_{1,1,1}^{C,in}$	6909.715171519039
$z_{cu,2}$	1.0	$RecLMTD_{cu,2}$	0.009589150478907727	$b_{1,2,2}^{C,in}$	4550.0
$z_{hu,1}$	1.0	$RecLMTD_{hu,1}$	0.02257072000538577	$b_{2,1,2}^{C,in}$	4162.259084595902
$A_{1,1,1}^\beta$	70.76867866839005	$q_{1,1,1}$	676.918555846366	$b_{1,1,1}^{C,out}$	7586.633727365405
$A_{1,2,2}^\beta$	68.57240847861726	$q_{1,2,2}$	1950.0	$b_{1,2,2}^{C,out}$	6500.0
$A_{2,1,2}^\beta$	142.98255202451594	$q_{2,1,2}$	2434.6221662304524	$b_{2,1,2}^{C,out}$	6596.881250826354
$A_{cu,1}^\beta$	5.030173826900646	$q_{cu,1}$	173.0814441536338		
$A_{cu,2}^\beta$	37.68654427850372	$q_{cu,2}$	1965.3778337695476		
$A_{hu,1}^\beta$	13.224123753786348	$q_{hu,1}$	488.4592779231816		
$A_{1,1,1}$	70.76867866839005	$t_{1,1}^{(H)}$	650.0		
$A_{1,2,2}$	68.57240847861726	$t_{1,2}^{(H)}$	582.3081444153634		
$A_{2,1,2}$	142.98255202451594	$t_{1,3}^{(H)}$	387.30814441536336		
$A_{cu,1}$	5.030173826900646	$t_{2,1}^{(H)}$	590.0		
$A_{cu,2}$	37.68654427850372	$t_{2,2}^{(H)}$	590.0		
$A_{hu,1}$	13.224123753786348	$t_{2,3}^{(H)}$	468.26889168847737		
$dt_{1,1,1}$	32.56395186154544	$t_{1,1}^{(C)}$	617.4360481384546		
$dt_{1,1,2}$	10.0	$t_{1,2}^{(C)}$	572.3081444153635		
$dt_{1,2,2}$	82.30814441536347	$t_{1,3}^{(C)}$	410.0		
$dt_{1,2,3}$	37.30814441536339	$t_{2,1}^{(C)}$	500.0		
$dt_{2,1,2}$	17.691855584636528	$t_{2,2}^{(C)}$	500.0		
$dt_{2,1,3}$	58.268891688477396	$t_{2,3}^{(C)}$	350.0		
$dt_{cu,1}$	67.30814441536339	$t_{1,1,1}^H$	370.0		
$dt_{cu,2}$	148.26889168847737	$t_{1,2,2}^H$	370.0		
$dt_{hu,1}$	62.56395186154544	$t_{2,1,2}^H$	468.26889168847737		
$f_{1,1,1}^H$	2.4175662708798784	$t_{1,1,1}^C$	628.394494805876		
$f_{1,2,2}^H$	9.185036271974024	$t_{1,2,2}^C$	500.0		
$f_{2,1,2}^H$	20.0	$t_{2,1,2}^C$	649.7011325580611		
$f_{1,1,1}^C$	12.07304295323127	$b_{1,1,1}^{H,in}$	1571.418076071921		
$f_{1,2,2}^C$	13.0	$b_{1,2,2}^{H,in}$	5348.463420630389		
$f_{2,1,2}^C$	10.151851425843663	$b_{2,1,2}^{H,in}$	11800.0		

3 *RecLMTD* Tangent Points

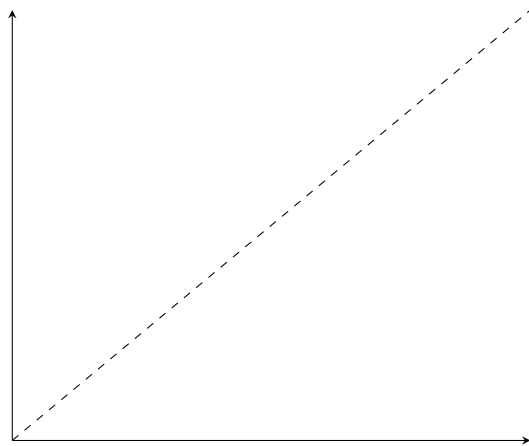
- $(1, 1, 1)$
 - Value: 0.052323
 - Error: 0.000000012148853
 - Relative error: 0.000000232188130
- $(1, 2, 2)$
 - Value: 0.017583
 - Error: 0.000000851406524
 - Relative error: 0.000048420709420
- $(2, 1, 2)$
 - Value: 0.029375
 - Error: 0.000000162096492
 - Relative error: 0.000005518115809
- $cu, 1$
 - Value: 0.014568
 - Error: 0.000000005368673
 - Relative error: 0.000000368534058
- $cu, 2$
 - Value: 0.009589
 - Error: 0.000000000495470
 - Relative error: 0.000000051669828
- $hu, 1$
 - Value: 0.022571
 - Error: 0.000000001289351
 - Relative error: 0.000000057124932



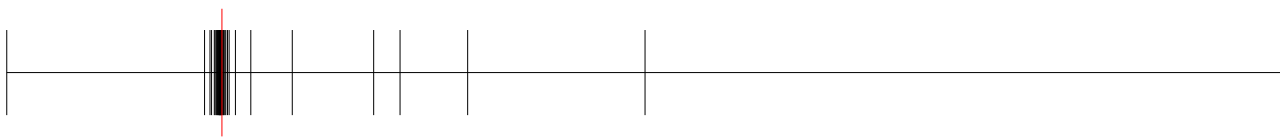
(a) *RecLMTD* - (1, 1, 1)



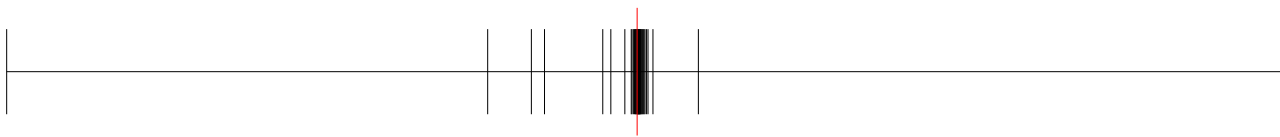
(b) *RecLMTD* - (1, 2, 2)



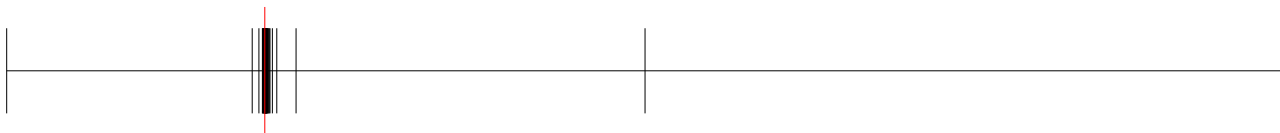
(c) *RecLMTD* - (2, 1, 2)



(a) *RecLMTD* - cu, 1



(b) *RecLMTD* - cu, 2



(a) *RecLMTD* - hu, 1

4 Balancing Breakpoints

4.1 $b^{H,\text{in}}$

- (1, 1, 1)
 - Value: 1571.418076071921
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 5348.463420630389
 - Absolute error: 0.05800729061138554
 - Relative error: -1.0845481577126876e-05
- (2, 1, 2)
 - Value: 11800.0
 - Absolute error: 0.0
 - Relative error: 0.0

4.2 $b^{H,\text{out}}$

- (1, 1, 1)
 - Value: 894.499520225555
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 3398.463420630389
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 9365.377833769548
 - Absolute error: 0.0
 - Relative error: 0.0

4.3 $b^{C,\text{in}}$

- (1, 1, 1)
 - Value: 6909.715171519039
 - Absolute error: 0.2143615082704855
 - Relative error: 3.102416718150025e-05
- (1, 2, 2)
 - Value: 4550.0
 - Absolute error: 0.0

- Relative error: 0.0
- (2, 1, 2)
 - Value: 4162.259084595902
 - Absolute error: 0.0
 - Relative error: 0.0

4.4 $b^{C,out}$

- (1, 1, 1)
 - Value: 7586.633727365405
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 6500.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 6596.881250826354
 - Absolute error: 1.2118818945582461
 - Relative error: 0.00018373903038055363



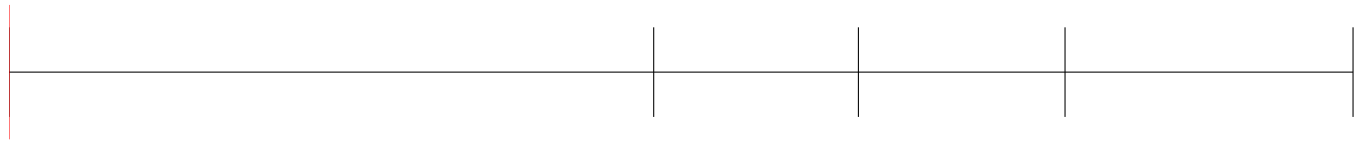
(a) $t^{(H)} - (1, 1)$



(a) $t^{(H)} - (1, 2)$



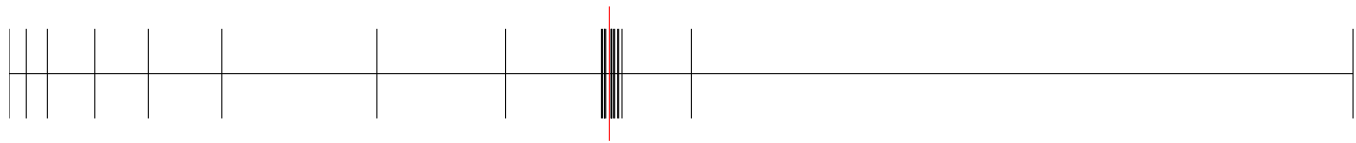
(a) $t^{(H)} - (2, 2)$



(a) $t^{(H)} - (1, 1, 1)$



(a) $t^{(H)} - (1, 2, 2)$



(a) $t^{(H)} - (2, 1, 2)$



(a) $t^{(C)} - (1, 2)$



(a) $t^{(C)} - (2, 3)$

5 Area Breakpoints

5.1 A

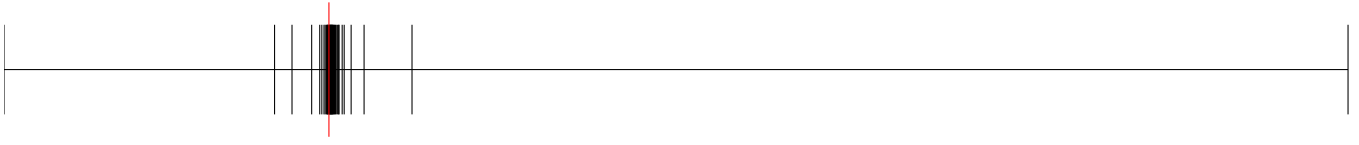
- $(1, 1, 1)$
 - Value: 70.768679
 - Error: 0.068553630707981
 - Relative error: 0.000967762693191
- $(1, 2, 2)$
 - Value: 68.572408
 - Error: 0.0000000000000057
 - Relative error: 0.0000000000000001
- $(2, 1, 2)$
 - Value: 142.982552
 - Error: 0.052334304953632
 - Relative error: 0.000365884899108

5.2 A_{CU}

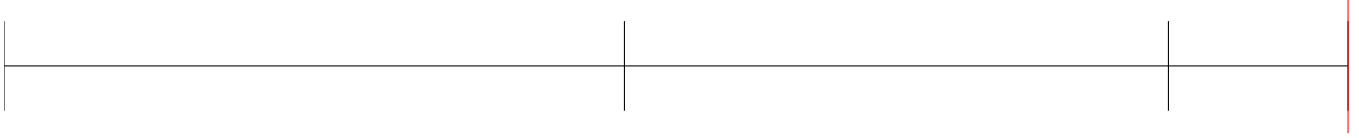
- $cu, 1$
 - Value: 5.030174
 - Error: 0.012602332174224
 - Relative error: 0.002499086173307
- $cu, 2$
 - Value: 37.686544
 - Error: 0.006063313348058
 - Relative error: 0.000160862135454

5.3 A_{HU}

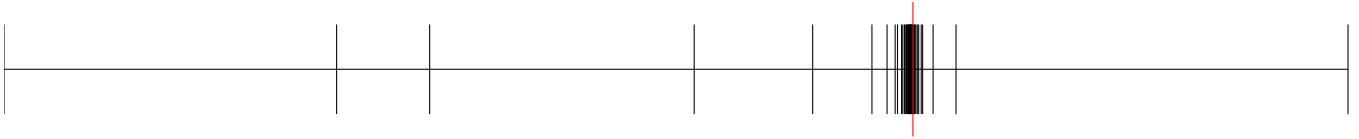
- $hu, 1$
 - Value: 13.224124
 - Error: 0.005729361458103
 - Relative error: 0.000433063119310



(a) $A^\beta - (1, 1, 1)$



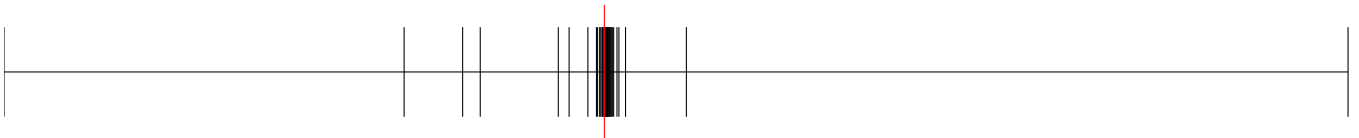
(a) $A^\beta - (1, 2, 2)$



(a) $A^\beta - (2, 1, 2)$



(a) $A^\beta - cu, 1$



(a) $A^\beta - cu, 2$

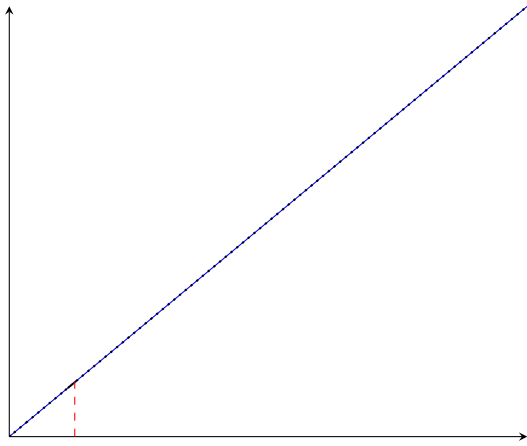


(a) $A^\beta - hu, 1$

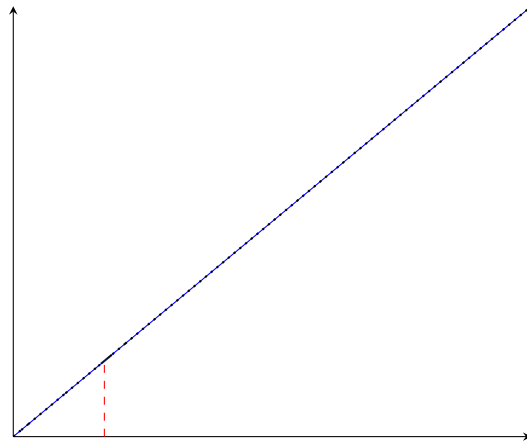
6 A^β Breakpoints

- (1, 1, 1)
 - q : 70.768679
 - $\hat{q}^\beta$: 70.768679
 - q^β : 70.768679
 - Breakpoints: [0, 5.0, 18.13353447520507, 60.38493803732795, 63.3468696286352, 65.40595072874316, 67.27362048222933, 69.14107665413653, 69.42777867315647, 69.49758876855047, 69.89902342903967, 70.11420238741596, 70.22290217272871, 70.30181758862284, 70.46247894104941, 70.50425945122109, 70.58322282660124, 70.60541450524272, **70.65581083979713**, **70.83181816751056**, 70.8571173980136, 70.94547904612048, 70.95870595008942, 71.00935653192329, 71.03465240716494, 71.1226084153824, 71.18527282143235, 71.21162487917152, 71.22933662920877, 71.2616536557244, 71.33750386361461, 71.3743618177527, 71.62767096549581, 71.6637831223933, 71.88123217857378, 72.01156079815767, 72.36161285231599, 560.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (1, 2, 2)
 - q : 68.572408
 - $\hat{q}^\beta$: 68.572408
 - q^β : 68.572408
 - Breakpoints: [0, 6.000977230027827, 11.262728503004022, 39.36524453694061, 64.01825678947372, 66.17325582523148, 66.74637189255778, 67.27673099173637, 67.60071888621576, 67.91344539650363, 68.10969007738701, 68.20353230192663, **68.48329505760876**, **68.66676327863097**, 68.76047282726, 68.85281320938876, 68.94418284540035, 69.0314457108091, 69.13316279873226, 69.22655548799992, 69.3178258940278, 69.34502840170047, 69.41355730879985, 69.51025044804472, 69.60484346150977, 69.93657924014246, 70.08419437796795, 70.37078617087542, 70.56626459266522, 70.94081287188642, 71.26369181826138, 72.25348351595485, 72.87506706799363, 74.6496383881616, 84.1468297226002, 390.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (2, 1, 2)
 - q : 142.982552
 - $\hat{q}^\beta$: 142.982552
 - q^β : 142.982552
 - Breakpoints: [0, 9.898970403313276, 28.333333333333332, 42.69900440382105, 48.67452649027746, 79.83702775273511, 87.16107999267206, 118.88765652962957, 127.07690270456857, 128.52021081477858, 131.69760457629607, 132.12228195956155, 132.5928424015674, 135.25996987589937, 135.81963163113545, 137.02923097052596, 137.63821724432228, 138.86359281280824, 139.27702937706263, 139.47623233772285, 140.0019654200325, 140.31685261310218, 140.7008934199032, 140.73764915547463, 141.47857504067568, 141.79783607898793, 142.2268253113294, **142.4338700260237**, **143.30274961189664**, 144.42095395765313, 144.67005099128556, 144.70843093515643, 147.02575214945074, 147.07009017600456, 147.48374761404892, 149.84689443310864, 150.84969099235698, 720.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $cu, 1$
 - q : 5.030174
 - $\hat{q}^\beta$: 5.030174

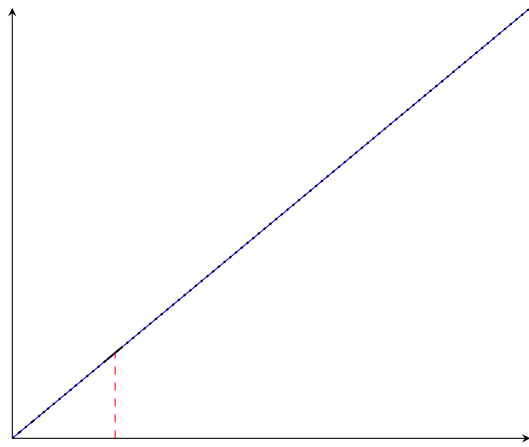
- q^β : 5.030174
- Breakpoints: [0, 2.5450346292088537, 2.9819181007907165, 3.1809419553840232, 3.9948823402911255, 4.0691182452509045, 4.269280092716329, 4.432280503220532, 4.440201431981878, 4.600843803756838, 4.613047202566896, 4.707108999269609, 4.75409580038526, 4.813467676447643, 4.825601156120121, 4.844843256177314, 4.849719301830286, 4.882089573558703, 4.888121952381175, 4.9192700692457025, 4.956260392645724, 4.962276317638335, 4.974201802143407, 4.983938626310074, 4.993068227518925, **4.993351789275153**, **5.0359974771272995**, 5.0418124100090695, 5.081618324442152, 5.0892956597199035, 5.195781532076625, 5.3386847474643435, 7.702824927089129, 9.28003023700696, 13.56934656224007, 14.535545040173215, 181.61828057849593]
- Error: 0.0000000000000000
- Relative error: 0.0000000000000000
- $cu, 2$
 - q : 37.686544
 - $\hat{q}^\beta$: 37.686544
 - q^β : 37.686544
 - Breakpoints: [0, 18.023935870161154, 21.04660840930274, 24.97364426355679, 30.189424467922265, 34.23465128608717, 36.670233878040335, 37.21616748032837, 37.40128299925594, 37.443498367229196, 37.47875108132638, 37.52720514437563, 37.554741031910254, 37.612770977560544, 37.63325504358228, 37.65435035996605, 37.6551299822246, **37.682669717644174**, **37.711317605988725**, 37.71812813120871, 37.738762662315665, 37.7631578546152, 37.76995088018704, 37.78573239590169, 37.791030068986316, 37.81486687126635, 37.821651654621526, 37.87287854392874, 37.88148462263016, 37.9018730666985, 37.9454585315849, 37.95085342166671, 38.035748486883605, 38.10872593446926, 38.21652191748154, 38.224970554765775, 38.88483572219945, 285.4001551947793]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $hu, 1$
 - q : 13.224124
 - $\hat{q}^\beta$: 13.224124
 - q^β : 13.224124
 - Breakpoints: [0, 4.943755299006499, 12.544341108707222, 12.595096067947068, 12.917705220652834, 13.064750414488246, 13.104034300095861, 13.134779411775817, 13.161020122653593, 13.189852638760229, 13.192252946941991, 13.195607617969486, 13.211215176722128, 13.213749911872785, **13.218222920629291**, **13.22717920580809**, 13.23798359102618, 13.238585544475612, 13.244985069124171, 13.251789114535772, 13.265751604070672, 13.266498143734903, 13.269097009439808, 13.279580305153486, 13.293477718720517, 13.294574825457861, 13.30871858348944, 13.321736298873567, 13.32330648750218, 13.32663103908035, 13.346967780023082, 13.363827419897227, 13.411355634100326, 13.439500729482463, 13.468285806082731, 13.498843422507207, 14.09178614751222, 237.3002543523117]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000



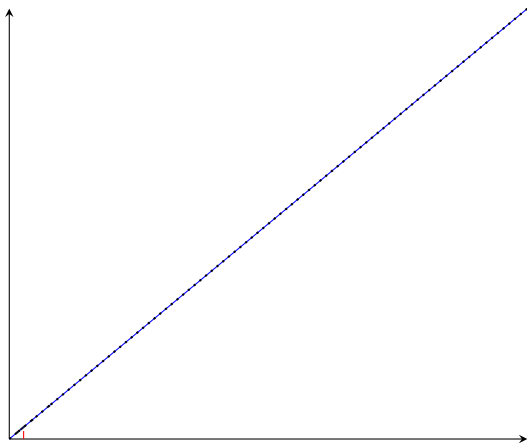
(a) $q^\beta - (1, 1, 1)$



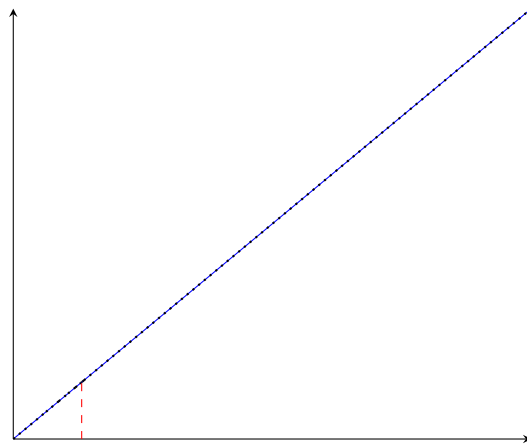
(b) $q^\beta - (1, 2, 2)$



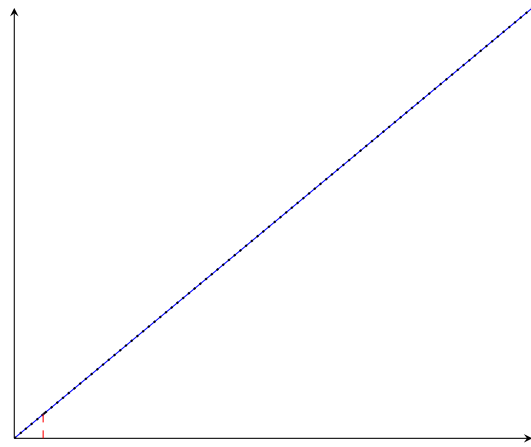
(c) $q^\beta - (2, 1, 2)$



(a) $q^\beta - cu, 1$



(b) $q^\beta - cu, 2$



(a) $q^\beta - hu, 1$

7 Other variables

Variable	Value	Variable	Value	Variable	Value
$z_{1,1,2}$	-0.0	$f_{2,2,1}^C$	0.0	$b_{2,2,2}^{H,out}$	0.0
$z_{1,2,1}$	-0.0	$f_{2,2,2}^C$	0.0	$b_{1,1,2}^{C,in}$	1987.7409154040981
$z_{2,1,1}$	-0.0	$RecLMTD_{1,1,2}$	0.013817625349338898	$b_{1,2,1}^{C,in}$	6500.0
$z_{2,2,1}$	-0.0	$RecLMTD_{1,2,1}$	0.0080685089611787	$b_{2,1,1}^{C,in}$	1674.9069947114122
$z_{2,2,2}$	-0.0	$RecLMTD_{2,1,1}$	0.029166666666666688	$b_{2,2,1}^{C,in}$	0.0
$z_{hu,2}$	-0.0	$RecLMTD_{2,2,1}$	0.020111728838038176	$b_{2,2,2}^{C,in}$	0.0
$A_{1,1,2}^\beta$	0.0	$RecLMTD_{2,2,2}$	0.022491280030922028	$b_{1,1,2}^{C,out}$	1987.7409154040988
$A_{1,2,1}^\beta$	0.0	$RecLMTD_{hu,2}$	0.005549329826813255	$b_{1,2,1}^{C,out}$	6500.0
$A_{2,1,1}^\beta$	0.0	$q_{1,1,2}$	0.0	$b_{2,1,1}^{C,out}$	1674.9069947114122
$A_{2,2,1}^\beta$	0.0	$q_{1,2,1}$	0.0	$b_{2,2,1}^{C,out}$	0.0
$A_{2,2,2}^\beta$	0.0	$q_{2,1,1}$	0.0	$b_{2,2,2}^{C,out}$	0.0
$A_{hu,2}^\beta$	0.0	$q_{2,2,1}$	0.0		
$A_{1,1,2}$	0.0	$q_{2,2,2}$	0.0		
$A_{1,2,1}$	0.0	$q_{hu,2}$	0.0		
$A_{2,1,1}$	0.0	$t_{1,1,2}^H$	567.6967965288007		
$A_{2,2,1}$	0.0	$t_{1,2,1}^H$	650.0		
$A_{2,2,2}$	0.0	$t_{2,1,1}^H$	590.0		
$A_{hu,2}$	0.0	$t_{2,2,1}^H$	590.0		
$dt_{1,1,3}$	240.0	$t_{2,2,2}^H$	370.0		
$dt_{1,2,1}$	150.0	$t_{1,1,2}^C$	410.0		
$dt_{2,1,1}$	16.47481108203014	$t_{1,2,1}^C$	500.0		
$dt_{2,2,1}$	90.0	$t_{2,1,1}^C$	565.7340298780514		
$dt_{2,2,2}$	10.0	$t_{2,2,1}^C$	350.0		
$dt_{2,2,3}$	33.29094213760047	$t_{2,2,2}^C$	440.01120358475225		
$dt_{hu,2}$	180.0	$b_{1,1,2}^{H,in}$	474.61802352324503		
$f_{1,1,2}^H$	0.8149637280259764	$b_{1,2,1}^{H,in}$	4928.581923928079		
$f_{1,2,1}^H$	7.582433729120122	$b_{2,1,1}^{H,in}$	11800.0		
$f_{2,1,1}^H$	20.0	$b_{2,2,1}^{H,in}$	0.0		
$f_{2,2,1}^H$	0.0	$b_{2,2,2}^{H,in}$	0.0		
$f_{2,2,2}^H$	0.0	$b_{1,1,2}^{H,out}$	474.61802352324503		
$f_{1,1,2}^C$	4.848148574156337	$b_{1,2,1}^{H,out}$	4928.581923928078		
$f_{1,2,1}^C$	13.0	$b_{2,1,1}^{H,out}$	11800.0		
$f_{2,1,1}^C$	2.926957046768729	$b_{2,2,1}^{H,out}$	0.0		